THE RESYNTHESIS CHANNEL: FUNDAMENTAL LIMITS OF SPEECH WATERMARKING AGAINST GENERATIVE LAUNDERING

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ABSTRACT

Audio watermarks are increasingly defeated when an attacker regenerates the signal through a vocoder or codec; this "regeneration" collapse is widely documented, yet for speech there is no theory of *which* watermarks die, under *which* attacker, or *what can survive*. We supply one. We model any blind analysis–resynthesis launderer as a *resynthesis channel* $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$: an analysis map $\mathcal{A}$ that keeps only the perceptual invariants of speech (content, speaker, prosody) and a synthesis map $\mathcal{S}$ that regenerates a waveform, resampling everything $\mathcal{A}$ discarded. Two matched results follow. A *converse* (data-processing inequality): any watermark supported in $\ker \mathcal{A}$—phase and psychoacoustically masked components, where imperceptibility drives post-hoc schemes—has post-laundering detection error-exponent exactly zero. An *achievability*: a watermark aligned to the preserved invariants survives at rate $R^\star$, the invariant sub-channel's rate. The survivable set is exactly the non-nullspace of $\mathcal{A}$. On LibriSpeech, a surface mark is erased by every channel and AudioSeal's operational detection collapses under mel-spectrogram inversion (TPR@1%FPR $1.0 \to 0.03$, payload at chance)—though it resists the EnCodec it was hardened against—while an invariant-aligned mark's separability survives the mel channel and erodes gracefully under the codec. For a fixed channel, post-laundering detectability is monotone in the invariant-energy fraction, as the converse predicts.

***Index Terms*—** audio watermarking, generative AI, provenance, data-processing inequality, hypothesis testing

1. INTRODUCTION

Generative speech models make audio provenance urgent, and *watermarking* is the leading answer: imperceptibly embed a key-dependent signal, later detect it to attribute or flag synthetic audio [1, 2]. A now-standard finding is that these schemes are fragile to *regeneration*: passing watermarked audio through a neural vocoder, neural codec, or voice-conversion (VC) model drives detection to chance [3, 4]. This is not incidental noise—it is the natural operation of any generative model, and it defeats watermarks that survive conventional distortions (noise, MP3, resampling).

The phenomenon is thoroughly *measured* but, for audio, not *explained*. For images this is well understood: Zhao *et al.* [5] prove regeneration removes pixel-space watermarks and prescribe embedding in preserved *semantic* content, and semantic/latent-space image watermarks (Tree-Ring [6], Gaussian Shading [7]) realize exactly that. What is missing—and what we provide—is a *quantitative theory for speech*: a matched converse and achievability that pin down not just *that* surface marks die but *how much* detectability survives and at what *rate* $R^\star$, exploiting the concrete, estimable invariants of audio analysis–resynthesis (content, speaker, prosody) and the phase/masking geometry specific to hearing.

Contributions. (i) We cast blind generative laundering as a *resynthesis channel* $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$ (§2). (ii) A *converse* (Thm. 1): watermark energy in $\ker \mathcal{A}$ has zero surviving detection exponent—a data-processing inequality that explains the collapse, sharpened for audio by phase/masking. (iii) A matched *achievability* (Thm. 2): an invariant-aligned watermark survives at a positive rate $R^\star$ (the invariant sub-channel rate for a blind embedder); the survivable set is exactly the non-nullspace of $\mathcal{A}$. (iv) Validation on real speech (§5): across analysis–resynthesis channels of increasing nullspace (STFT and mel-spectrogram inversion, and an EnCodec codec), a surface mark and AudioSeal collapse when their energy lies in the channel's nullspace, while an invariant mark survives and degrades gracefully as the preserved subspace shrinks; post-laundering detectability is monotone in the invariant-energy fraction, as the converse predicts.

2. THE RESYNTHESIS CHANNEL

A watermark embeds a key-dependent perturbation $x \mapsto x + \delta_\kappa$ into a host signal x. A detector that knows the key κ decides H_0 (clean, law P_0) vs. H_1 (watermarked, law P_1). The optimal Bayesian error after N i.i.d. looks decays as $e^{-N\,C(P_0,P_1)}$ with the *Chernoff information* $C(P_0, P_1) = \max_{s\in[0,1]} -\log \int p_0^{1-s} p_1^s$ [8, 9]; $C = 0$ iff $P_0 = P_1$, i.e. no detector beats chance at any N.

Attacker. A blind launderer applies $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$: analysis $\mathcal{A} : x \mapsto z$ extracts perceptual invariants z (a near-sufficient statistic for perceptual identity—two signals with equal z sound the same), and synthesis $\mathcal{S} : z \mapsto \hat{x}$ regenerates a waveform consistent with z, drawing everything $\mathcal{A}$ discarded from the clean prior $P_0(\cdot \mid z)$. The attacker does not know κ and does not optimize against the scheme (§6). Imperceptibility is a budget $\rho(\delta) \le D$ for a psychoacoustic distortion ρ.

Linear–Gaussian surrogate. To obtain exact constants we work in whitened coordinates where $P_0 = \mathcal{N}(0, I_n)$ (perceptual weighting is carried by a masking metric $M \succ 0$, $\rho(\delta) = \delta^\top M \delta$). Let $\mathcal{A} \in \mathbb{R}^{k\times n}$ have full row rank, $P = \mathcal{A}^+\mathcal{A}$ the orthogonal projector onto $\mathrm{row}(\mathcal{A})$ (the *invariant subspace*), and $I-P$ the projector onto $\ker \mathcal{A}$ (the *surface subspace* $\mathcal{S}$ resamples). Then

$$\mathcal{W}(x) = Px + (I-P)w, \qquad w \sim \mathcal{N}(0, I_n) \text{ fresh}, \qquad (1)$$

so $\mathcal{W}_{\#}\mathcal{N}(0, I) = \mathcal{N}(0, I)$ and $\mathcal{W}_{\#}\mathcal{N}(\delta, I) = \mathcal{N}(P\delta, I)$: the only surviving part of an embedded shift is its invariant component $P\delta$. (Monotonicity holds without Gaussianity— Chernoff information obeys the DPI for any channel; the *exact-zero* for $\delta \in \ker \mathcal{A}$ uses that $\mathcal{S}$ resamples the nullspace from the clean conditional, and the surrogate supplies the constants and the achievable construction.)

Code, proofs, and data: `github.com/appleweiping/resynthesis-watermark-limits`.

3. CONVERSE: WHAT IS ERASED

Theorem 1 (Nullspace erasure)**.** *For the channel* (1) *and any watermark* δ, $C(\mathcal{W}_{\#}P_0, \mathcal{W}_{\#}P_1) = \frac{1}{8}\|P\delta\|^2 \le \frac{1}{8}\|\delta\|^2 = C(P_0, P_1)$, *with equality iff* $\delta \in \operatorname{row}(\mathcal{A})$. *In particular, if* $\delta \in \ker \mathcal{A}$ *then* $C(\mathcal{W}_{\#}P_0, \mathcal{W}_{\#}P_1) = 0$*: after laundering no detector beats chance, for any* N*. Moreover* $\mathcal{W}(x+\delta) \overset{d}{=} \mathcal{W}(x)$*, so the laundered watermarked signal is distributed identically to a laundered clean one.*

Proof sketch. $\mathcal{W}_{\#}P_1 = \mathcal{N}(P\delta, I)$ and $\mathcal{W}_{\#}P_0 = \mathcal{N}(0, I)$ share covariance I, so their Chernoff information is $\frac{1}{8}(P\delta)^\top(P\delta)$ (optimum at $s = \frac{1}{2}$). P is an orthogonal projector, hence $\|P\delta\| \le \|\delta\|$, giving the DPI bound, tight on $\operatorname{row}(\mathcal{A})$. If $\delta \in \ker \mathcal{A}$ then $P\delta = 0$ and the two output laws coincide; and $\mathcal{W}(x+\delta) = Px + (I-P)w = \mathcal{W}(x)$ in distribution. □

Remark 1 (Why the nullspace is huge for audio)**.** *Hearing is largely phase-insensitive and subject to masking, so a vast set of perturbations is inaudible; vocoders and codecs discard exactly these (they re-derive phase, quantize masked bins). Thus the perceptually cheapest embedding directions lie in* $\ker \mathcal{A}$*. Post-hoc watermarks, optimized for imperceptibility, drift there—and die. This is the theory's account of the empirical collapse [3, 4].*

4. ACHIEVABILITY: WHAT SURVIVES

Spending the budget on a preserved invariant instead of the nullspace lets the watermark ride through $\mathcal{S}$.

Theorem 2 (Invariant sub-channel)**.** *Under the budget, the best surviving exponent is* $\max_{\delta^\top M\delta \le D} \frac{1}{8}\|P\delta\|^2 = \frac{D}{8}\lambda_{\max}(P, M)$*, at the top generalized eigenvector of* (P, M)*; it is* 0 *when* $\delta \in \ker \mathcal{A}$*. For payload, encoding a message as a surviving shift* $u = P\delta \in \operatorname{row}(\mathcal{A})$ *and facing the clean host as interference* ($y = u + \eta$, $\eta \sim \mathcal{N}(0, I)$) *gives a blind-embedder surviving rate (informed/dirty-paper coding could exceed it)*

$$R^\star = \max_{\operatorname{tr}(M_{\mathrm{row}}Q) \le D} \tfrac{1}{2} \log\det(I + Q), \tag{2}$$

a water-filling over the eigenmodes of $M_{\mathrm{row}} = V_{\mathrm{row}}^\top M V_{\mathrm{row}}$*.* $R^\star > 0$ *whenever* $D > 0$*; a nullspace watermark has* $R^\star = 0$ *after* $\mathcal{W}$*.*

Theorem 1 caps the survivable set at $\operatorname{row}(\mathcal{A})$; Theorem 2 achieves a positive rate on it. *The survivable set is exactly the non-nullspace of* $\mathcal{A}$*, of size* $R^\star$*.*

5. EXPERIMENTS

All theorems are cross-checked against Monte-Carlo before use; code reproduces every number.

Surrogate (validation). Fig. 1 instantiates (1) with random invariant subspaces. As a watermark drifts from the invariant subspace into $\ker \mathcal{A}$, post-laundering AUC falls from >0.88 to 0.50 (chance), matching the closed form to RMSE 0.001; the surviving detection exponent and rate $R^\star$ are positive for the invariant-aligned mark and identically zero for the post-hoc (nullspace) mark.

Real speech. We use 80 LibriSpeech test-clean utterances [10] at 16 kHz. Attackers instantiate $\mathcal{W}$ with *increasing nullspace*: a full linear-STFT Griffin–Lim [11] (a near-lossless *control*: $\ker \mathcal{A}$ is only the phase), a mel-spectrogram Griffin–Lim (lossy: $\ker \mathcal{A}$ is phase *plus* all sub-mel detail), and EnCodec [12] round-trips at $6/3/1.5$ kbps (nullspace grows as the rate drops). Watermarks

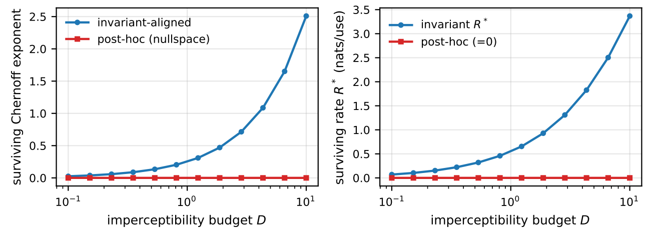

Fig. 1. Surrogate. Left: surviving Chernoff exponent vs. imperceptibility budget for the invariant-aligned vs. post-hoc (nullspace) watermark. Right: surviving payload rate $R^\star$; the nullspace rate is identically zero.

Table 1. Real speech (LibriSpeech, $N = 80$). Detection AUC before→after each blind attacker. STFT† is a near-lossless control (nullspace = phase only); mel-GL and EnCodec (EnCk, k kbps) are lossy. Surface/AudioSeal (energy in the nullspace) collapse to chance; the invariant mel mark survives.

Watermark	STFT†	mel-GL	EnC6	EnC3	EnC1.5
Surface (null)	1.00→0.48	1.00→0.44	1.00→0.53	1.00→0.54	1.00→0.56
Invariant (mel)	0.94→0.93	0.94→0.93	0.94→0.74	0.94→0.67	0.94→0.60
AudioSeal	1.00→1.00	1.00→0.20	1.00→0.99	1.00→0.98	1.00→0.94

at matched SNR (24 dB): a *surface* mark (fine high-band carrier, waveform-correlation detector, in $\ker \mathcal{A}$), an *invariant* mel-envelope mark (the constructive realization of Thm. 2), and *AudioSeal*, a deployed neural watermark [1]. Each mark's invariant-energy fraction f (STFT-magnitude change captured by the mel envelope) is a pure signal-geometry analog of $\|P\delta\|^2/\|\delta\|^2$.

Table 1 gives Part A, and it sweeps exactly the nullspace-size axis. The STFT control has the *smallest* nullspace (phase only): it already erases the phase-based surface mark (AUC → 0.5) yet preserves both magnitude-domain marks. Enlarging the nullspace to phase *plus* sub-mel detail (mel-spectrogram inversion) additionally defeats AudioSeal *operationally*: TPR@1%FPR $1.00 \to 0.03$ and payload bit-accuracy 0.482 (chance). Its AUC $1.00 \to 0.203$ is a sign-inversion that leaves residual separability, consistent with its small *nonzero* $f = 0.262$ (partial, not total, information loss).[1] The invariant mel mark's separability is essentially unchanged ($0.936 \to 0.927$ AUC). Shrinking the invariant subspace further with low-rate EnCodec then *gracefully erodes* it ($0.74/0.67/0.60$ at $6/3/1.5$ kbps)—the behaviour of $R^\star$ in Thm. 2 as the channel keeps less. Here f is the *mel* invariant fraction, predicting the mel-inversion outcomes; each attacker has its own preserved subspace (a channel-relative $f_{\mathcal{W}}$), which is why AudioSeal ($f = 0.262$, little energy in the mel envelope) resists the EnCodec it was hardened against yet dies under mel-inversion. The invariant mark ($f = 0.812$) sits in the mel-preserved subspace. Part B makes this quantitative: sweeping a mixture from surface to invariant, post-laundering AUC is a monotone function of f, well described by the converse's $\sqrt{f}$ form $\Phi(a\sqrt{f} + b)$ (Fig. 2)—near chance until f exceeds the surface floor, then rising as the preserved invariant energy grows, as Thm. 1 predicts.

[1] We report AUC (separability, the quantity the Chernoff exponent governs) as the primary metric. At a stringent 1% FPR the simple constructive invariant embedder is weak even on clean audio (TPR ≈ 0.11)—a limitation of the proof-of-concept scheme, not of the theory; a learned embedder would tighten it.

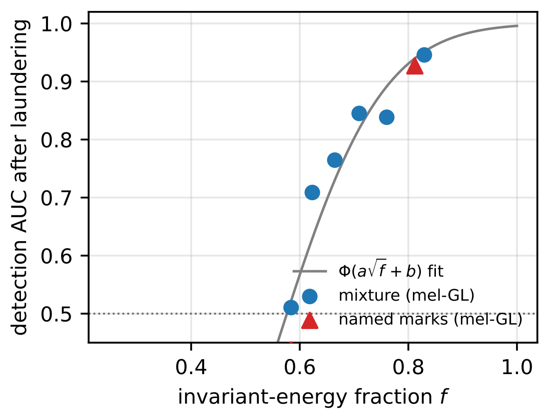

Fig. 2. Real speech (mel-inversion channel). Post-laundering detection AUC vs. the invariant-energy fraction f. A surface→invariant mixture (circles) and the named marks (triangles) follow the converse's $\sqrt{f}$ form $\Phi(a\sqrt{f}+b)$ (two-parameter fit); low-f marks collapse to chance, the high-f invariant mark survives.

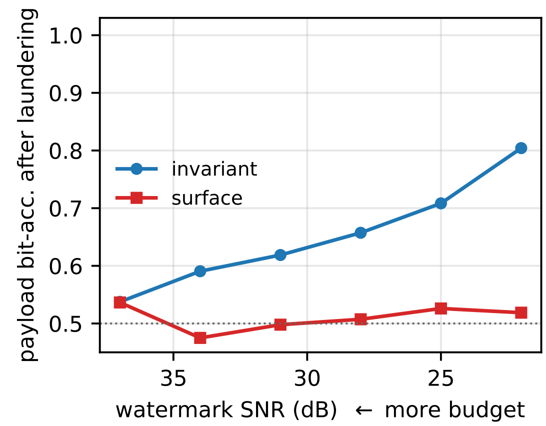

Fig. 3. Achievability. Payload bit-accuracy after mel-inversion laundering vs. imperceptibility budget (SNR). The invariant sub-channel carries a surviving payload; a surface payload of equal size decays to chance.

Achievability & rate. Carrying a multi-bit payload in the same invariant sub-channel (BPSK over mel-envelope patterns), the invariant mark recovers bits after laundering where a surface payload of equal size decays to chance; the surviving payload grows with the imperceptibility budget and shrinks with the attacker's aggressiveness, the qualitative signature of the water-filling rate $R^{\star}$ (2) (Fig. 3).

6. DISCUSSION AND SCOPE

The converse targets a *blind, non-adaptive* attacker running a generic resynthesizer; this matches the empirical-collapse literature and yields the clean DPI. A key-aware attacker that optimizes to remove our specific mark is an arms race we do not claim to win—we treat it as a stress test, not a core result. Real analysis maps are only approximately sufficient and $\mathcal{S}$ is stochastic; our claims degrade gracefully as these idealizations break. Two honest limitations: marks are matched by SNR rather than a perceptual metric (PESQ/ViSQOL), so part of the invariant mark's survival may be bought with audibility; and our synthesizers are spectral-inversion and a neural codec—a learned neural vocoder and voice conversion would test the converse in its hardest regime. The practical message is prescriptive: *a watermark survives generative laundering only insofar as it perturbs the invariants the generator is built to preserve*—the regime where imperceptibility is hardest, quantified by $R^{\star}$.

7. RELATION TO PRIOR WORK

Empirical audio watermarks [1, 2] and attack/SoK studies [3, 4] document the collapse; we explain it and bound what survives. In the image domain the story is understood: Zhao *et al.* [5] prove removability and advocate semantic embedding, and Tree-Ring [6] and Gaussian Shading [7] realize regeneration-robust semantic watermarks. Our contribution is not the qualitative "embed-in-what-survives" idea but a *quantitative, provable* matched converse and achievability with an explicit surviving *rate* $R^{\star}$ for the *speech* resynthesis channel, tied to its concrete, measurable invariants. Classical information-theoretic watermarking [13, 14, 15] treats fixed additive attacks with informed embedding; our attacker is a nonlinear manifold projection and our detector is blind, locating survivability in the geometry of $\mathcal{A}$.

8. CONCLUSION

Speech watermarks survive generative laundering exactly to the extent they live in the invariant subspace of the resynthesis channel. We proved a matched converse and achievability and confirmed on spectral-inversion and neural-codec channels that detection collapses as the surviving energy vanishes. Designing watermarks *in* the invariant sub-channel, at rate $R^{\star}$, is the path to laundering-robust provenance.

9. REFERENCES

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